

Solving the Time-Dependent Schrödinger Equation

via the Crank–Nicolson Method: Stability, Unitarity & Efficient Linear Algebra

A complete mathematical derivation from operator formalism to tridiagonal linear systems and the Thomas algorithm

1. The Physical Problem

The **Time-Dependent Schrödinger Equation** (TDSE) governs quantum state evolution:

$$i\hbar \frac{\partial \Psi(x,t)}{\partial t} = \hat{H} \Psi(x,t) = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right] \Psi(x,t) \quad (1)$$

Formal solution - time-evolution operator:

$$\Psi(x, t+\Delta t) = \underbrace{e^{-i\hat{H}\Delta t/\hbar}}_{\hat{U}(\Delta t)} \Psi(x, t)$$

Unitarity: the Core Requirement	$\hat{H} = \hat{H}^\dagger$ (Hermitian) $\Rightarrow \hat{U}^\dagger \hat{U} = e^{+i\hat{H}\Delta t/\hbar} e^{-i\hat{H}\Delta t/\hbar} = \hat{I}$ This enforces probability conservation : $\int_{-\infty}^{+\infty} \Psi(x,t) ^2 dx = 1 \quad \forall t$ <i>Any valid numerical scheme must discretely replicate this.</i>
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The computational challenge: approximate $e^{-i\hat{H}\Delta t/\hbar}$ on a finite spatial grid while keeping the discrete propagator *exactly* unitary.

2. Space–Time Discretization

Introduce a uniform grid with $N+1$ spatial and M temporal points:

$$x_j = j \Delta x, \quad j = 0, \dots, N; \quad t_n = n \Delta t, \quad n = 0, 1, \dots$$

Central finite difference (2nd-order, $\mathcal{O}(\Delta x^2)$):

$$\left. \frac{\partial^2 \Psi}{\partial x^2} \right|_{x_j} \approx \frac{\Psi_{j+1}^n - 2\Psi_j^n + \Psi_{j-1}^n}{(\Delta x)^2} \quad (2)$$

Derived by Taylor expansion: $\Psi_{j\pm 1} = \Psi_j \pm \Delta x \Psi' + \frac{(\Delta x)^2}{2} \Psi'' \pm \frac{(\Delta x)^3}{6} \Psi''' + \mathcal{O}(\Delta x^4)$

The **discrete Hamiltonian** H_{FD} acts on the grid vector $\Psi^n = [\Psi_1^n, \dots, \Psi_N^n]^\top$:

$$(H_{FD} \Psi^n)_j = -\beta \Psi_{j-1}^n + (2\beta + V_j) \Psi_j^n - \beta \Psi_{j+1}^n, \quad \beta \equiv \frac{\hbar^2}{2m(\Delta x)^2}$$

Boundary Conditions	Dirichlet (infinite well): $\Psi_0^n = \Psi_N^n = 0$ Periodic (ring/lattice): $\Psi_0^n = \Psi_N^n, \Psi_{-1}^n = \Psi_N^{n-1}$ Absorbing (open boundary): Add $-i\Gamma(x)$ to $V(x)$ near edges
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3. Why Euler Schemes Fail

Stability test: substitute $\Psi_j^n = g^n e^{ikj\Delta x}$ (Fourier mode) into each scheme and compute amplification factor g .

Forward Euler (FTCS) $\Psi^{n+1} = (I - \frac{i\Delta t}{\hbar} H) \Psi^n$ $ g = \sqrt{1 + (\frac{\Delta t \lambda}{\hbar})^2} > 1$ Unconditionally unstable. Norm grows $\rightarrow$ blows up.	Backward Euler (BTCS) $\Psi^{n+1} = (I + \frac{i\Delta t}{\hbar} H)^{-1} \Psi^n$ $ g = \frac{1}{\sqrt{1 + (\Delta t \lambda / \hbar)^2}} < 1$ Stable, but artificially dissipative . Norm decays $\Rightarrow$ violates QM.	Crank–Nicolson Average of FTCS & BTCS $ g = \left \frac{1 - i\Delta t \lambda / (2\hbar)}{1 + i\Delta t \lambda / (2\hbar)} \right = 1$ Unconditionally stable. Exact norm conservation.
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Here λ denotes any eigenvalue of H_{FD} ; since H_{FD} is real-symmetric all eigenvalues are real, so $|g| = 1$ holds for *every* mode simultaneously - unconditional stability.

4. Padé Approximants & The Evolution Operator

The exact time-evolution operator is $\hat{U} = e^{-ix}$, where $x \equiv \hat{H}\Delta t/\hbar$. Direct computation of the matrix exponential $e^{-i\hat{H}\Delta t/\hbar}$ is impractically expensive. Instead, the Crank–Nicolson method employs the **Padé [1,1] approximant**:

$$e^{-ix} \approx R(x) = \frac{1 - ix/2}{1 + ix/2} \implies U_{CN} = \left(I + i\frac{\Delta t}{2\hbar} H \right)^{-1} \left(I - i\frac{\Delta t}{2\hbar} H \right) \quad (3)$$

Taylor Series Matching (Order of Accuracy)	Let's compare their expansions for a small time step x : Exact: $e^{-ix} = 1 - ix - \frac{x^2}{2} + i\frac{x^3}{6} + \mathcal{O}(x^4)$ Padé: $R(x) = \left(1 - i\frac{x}{2}\right) \left(1 - i\frac{x}{2} - \frac{x^2}{4} + i\frac{x^3}{8} + \dots\right)$ $= 1 - ix - \frac{x^2}{2} + i\frac{x^3}{4} + \mathcal{O}(x^4)$ The expressions match exactly up to $\mathcal{O}(x^2)$. The discrepancy begins at the $\mathcal{O}(x^3)$ term, meaning the local truncation error per step is $\mathcal{O}(\Delta t^3)$. This makes the method globally 2nd-order accurate in time.
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Unconditional Unitarity (Stability)	Why use a rational function instead of just truncating the Taylor series? For any real x (since $\hat{H}$ is Hermitian, its eigenvalues are strictly real), the magnitude of the approximation is: $R(x) ^2 = R^*(x)R(x) = \left(\frac{1 + ix/2}{1 - ix/2}\right) \left(\frac{1 - ix/2}{1 + ix/2}\right) \equiv 1$ While a truncated Taylor polynomial will eventually drift off the unit circle (causing solutions to blow up or decay), the Padé [1,1] approximant maps the entire real line perfectly onto the unit circle in the complex plane. This mathematically guarantees that probability is exactly conserved regardless of the size of Δt .
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5. Tridiagonal Matrix Structure of A and B

Let $r_j \equiv \gamma(2\beta + V_j)$ (diagonal) and $s \equiv \gamma\beta$ (off-diagonal). The linear system (4) reads:

$$\underbrace{\begin{pmatrix} 1+ir_1 & -is & 0 & \cdots & 0 \\ -is & 1+ir_2 & -is & \cdots & 0 \\ 0 & -is & 1+ir_3 & \cdots & 0 \\ \vdots & & & \ddots & -is \\ 0 & 0 & \cdots & -is & 1+ir_{N-1} \end{pmatrix}}_A \Psi^{n+1} = \underbrace{\begin{pmatrix} 1-ir_1 & +is & 0 & \cdots & 0 \\ +is & 1-ir_2 & +is & \cdots & 0 \\ 0 & +is & 1-ir_3 & \cdots & 0 \\ \vdots & & & \ddots & +is \\ 0 & 0 & \cdots & +is & 1-ir_{N-1} \end{pmatrix}}_B \Psi^n \quad (4)$$



Sparsity pattern of A (identical structure for B; signs flipped)

6. Thomas Algorithm - $\mathcal{O}(N)$ Tridiagonal Solver

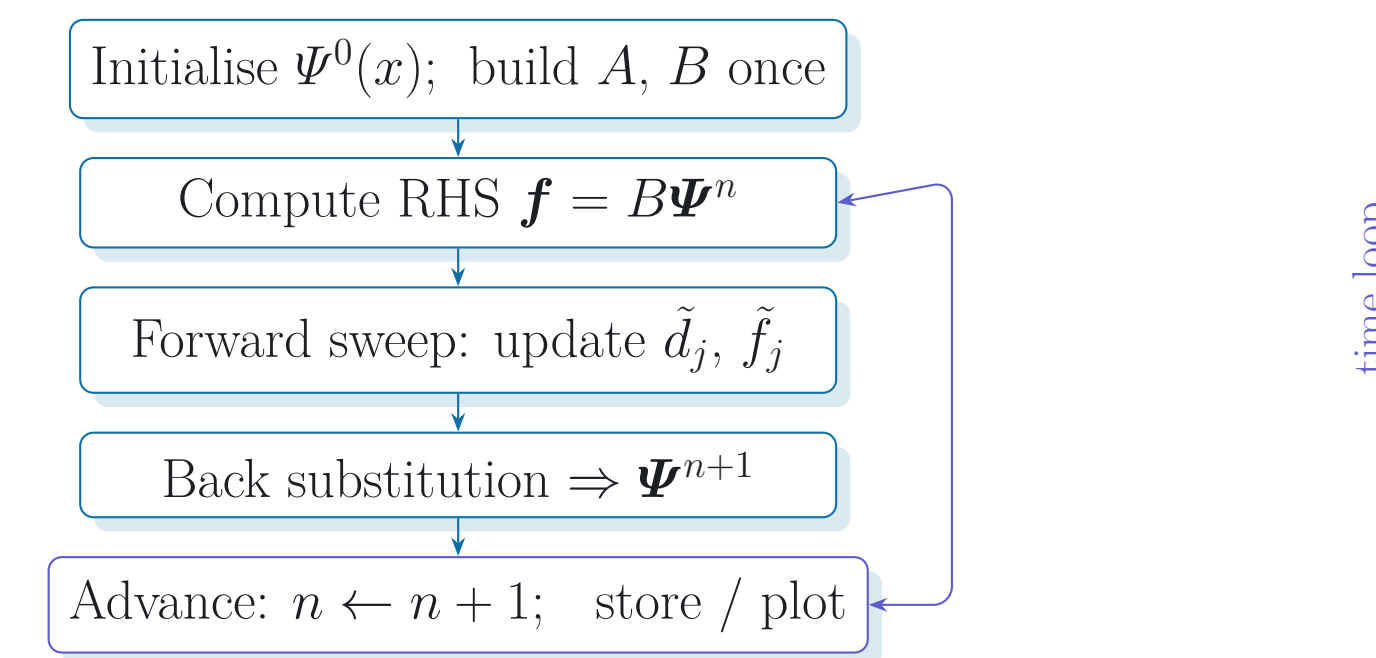
Naïve LU decomposition costs $\mathcal{O}(N^3)$. The Thomas algorithm exploits the **tridiagonal structure** to achieve $\mathcal{O}(N)$ with only $\approx 8N$ floating-point operations - no fill-in.

Let the system be $a_j \Psi_{j-1} + d_j \Psi_j + c_j \Psi_{j+1} = f_j$ with $a_j = c_j = -is$, $d_j = 1 + ir_j$.

Step 1 - Forward Sweep (modify coefficients)	Initialise: $\tilde{d}_1 = d_1, \tilde{f}_1 = f_1$. For $j = 2, 3, \dots, N-1$: $m_j = \frac{a_j}{\tilde{d}_{j-1}}, \quad \tilde{d}_j = d_j - m_j c_{j-1}, \quad \tilde{f}_j = f_j - m_j \tilde{f}_{j-1}$ Eliminates the sub-diagonal in $\mathcal{O}(N)$ steps.
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Step 2 - Back Substitution (recover Ψ^{n+1})	$\Psi_{N-1}^{n+1} = \frac{\tilde{f}_{N-1}}{\tilde{d}_{N-1}}, \quad \Psi_j^{n+1} = \frac{\tilde{f}_j - c_j \Psi_{j+1}^{n+1}}{\tilde{d}_j}, \quad j = N-2, \dots, 1$
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Full time-stepping algorithm:



7. Method Comparison at a Glance

Scheme	Stability	Norm	Accuracy	Cost/step
Forward Euler (FTCS)	Unstable	Grows	$\mathcal{O}(\Delta t, \Delta x^2)$	$\mathcal{O}(N)$
Backward Euler (BTCS)	Stable	Decays	$\mathcal{O}(\Delta t, \Delta x^2)$	$\mathcal{O}(N)$
Crank–Nicolson	Stable	Exact	$\mathcal{O}(\Delta t^2, \Delta x^2)$	$\mathcal{O}(N)$
Dense LU solve	Stable	Exact	$\mathcal{O}(\Delta t^2, \Delta x^2)$	$\mathcal{O}(N^3)$

CN with Thomas algorithm delivers the best possible combination: $\mathcal{O}(N)$ cost, $\mathcal{O}(\Delta t^2)$ accuracy, and exact unitarity - simultaneously.

8. Summary & Physical Significance

Complete Crank–Nicolson Recipe	Grid: choose $\Delta x, \Delta t$; set $\beta = \hbar^2/(2m\Delta x^2), \gamma = \Delta t/(2\hbar)$. Matrices: build tridiagonal $A = (1+i\gamma H_{FD})$ and $B = (1-i\gamma H_{FD})$ - <i>once</i>. Time loop: compute $f = B\Psi^n$; solve $A\Psi^{n+1} = f$ via Thomas Algorithm. Verify: $\ \Psi^{n+1}\ = 1$ to machine precision (guaranteed by U_{CN}).
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Unitarity Proof: $U_{CN} = (I + i\gamma H)(I - i\gamma H)^{-1}$ is a unitary Padé(1,1) approx.	Cayley form: $U_{CN} = (I + i\gamma H)(I - i\gamma H)^{-1}$ is a unitary Padé(1,1) approx.
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